

An exact matching characterization of the finite-dimensional projective tensor supremum of the I_{3322} Bell inequality

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Manuscript v0.1.2 — 19 September 2026

Abstract

We present a machine-assisted exact characterization of the supremum S_t of a specified normalization of I_{3322} , over binary projective measurements on arbitrary finite-dimensional tensor products, with no fixed bound on local dimension. A selected positive Jacobi carrier at the supremum has strictly decreasing labels and prescribed outer tails. Its stationary recurrence converts the problem into full-state matching of a one-dimensional unstable tail and a one-dimensional stable tail. A complete interval-certified analysis with two independent phases excludes asymmetric full matches before reducing to an explicitly defined analytic scalar residual. That residual has a strictly positive derivative throughout the full admissible parameter interval and exactly one zero. Consequently S_t is the unique matching parameter, with a strict rational enclosure of width $4.60932356071910243385112719549938850842 \times 10^{-52}$. The proof separates analytic arguments, finite certificates, and external theorem premises. It uses three specified interfaces from Coladangelo’s v1 preprint and the standard real-analytic stable/unstable manifold theorem. It makes no commuting-operator equality claim. The frozen proof survives internal reconstruction, exact replay, and adversarial and provenance audits; a non-load-bearing midpoint-helper defect remains disclosed. External expert human validation and journal peer review of this exact-value result have not yet occurred.

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1 Introduction

The I_{3322} Bell scenario has three binary measurements for each of two parties. Determining its quantum supremum is a concrete optimization problem in which the dimension of the underlying Hilbert spaces is unbounded, even when each strategy is finite-dimensional. Numerical optimization can suggest a limiting value, but does not identify that value exactly or establish that all admissible stationary configurations have been accounted for.

Pál and Vértesi studied high-dimensional strategies and the limiting-value problem [5]. Recent preprints by Pauwels [6] and Coladangelo [2] address the relation between finite-dimensional strategies and variational descriptions. Mghirbi's released rational witnesses [3, 4] provide finite objects from which a narrow interval can be checked exactly. Here the finite witnesses are independently verified, while the specific Coladangelo premises are stated in Section 3. Pál–Vértesi and Pauwels supply historical context and comparison, not premises of the exact-value theorem proved below. In particular, the theorem is not presented as a consequence of numerical agreement or of a formalization of a different statement.

Our target S_t , defined in Section 2, is the *finite-dimensional, projective tensor-product supremum*. The subscript t denotes this scope. It is unrelated to the outer-tail variable $t = a(s)$ used later. No commuting-operator supremum is identified with S_t , and no such identification is needed.

Theorem 1.1 (Exact-value characterization). *Let S_t be the supremum in Definition 2.1, and let*

$$L = 0.2508753845139765, \quad U = 0.2508753855, \quad I = (L, U].$$

Let $\mathcal{M}_s$ be the full-state matching set in Definition 5.3, with valid decreasing chains and crossing convention $x \geq 0 > y$. There is a unique $s_ \in I$ with $\mathcal{M}_{s_*} \neq \emptyset$, and*

$$\boxed{S_t = s_*}.$$

Equivalently, with the exactly specified analytic germ, propagated chart, implicit phase and residual of Section 6,

$$S_t = \sigma(t_*), \quad \mathcal{R}(t_*) = 0, \quad t_* \in a(I),$$

where t_ is the unique zero on that entire interval. The exact rational endpoints ℓ, h displayed below satisfy $\ell < S_t < h$. The conclusion uses External premises 3.2 and 5.1, together with the finite certificates explicitly specified in this manuscript.*

$$\ell = 0.250875384513976535617336610945951298901321973971476246796391973158622991323050603025506260$$

$$h = 0.250875384513976535617336610945951298901321973971476707728748045068866376435770152964357102$$

The two 90-place terminating decimals are rational numbers, and

$$\begin{aligned} h - \ell &= \frac{460932356071910243385112719549938850842}{10^{90}} \\ &= 4.60932356071910243385112719549938850842 \times 10^{-52}. \end{aligned} \tag{1.1}$$

The exact result is the unique-root characterization, rather than a claim that either decimal endpoint is the value or that an elementary closed form has been found.

The proof has three mathematical stages. First, a finite lower witness and a dimension-independent positive polynomial certificate put S_t in I . A variational compactness argument then produces one positive, square-summable, strictly decreasing stationary carrier at S_t , with outer endpoint limits. Second, hyperbolic local dynamics turns the carrier problem into matching two

tails in all three state coordinates. Third, a validated analytic chart and exhaustive finite covering argument treat the two independent tail phases. Only after the full matching equations exclude off-diagonal phases is a central scalar residual used. Its derivative has a strict sign throughout the full parameter interval. The exact identification is then a short theorem composition; high precision root isolation follows after uniqueness is established.

This is a machine-assisted proof with replayable certificates. The numerical subdivisions certify finitely many interval predicates; analytic arguments explain why those predicates imply statements about every admissible chain. The proof is not a proof-assistant formalization. The validation record is reported separately in Sections 9–11 and Appendix E, including the remaining implementation and provenance caveats and the absence of external expert human validation and journal peer review of the new exact-value theorem.

2 Bell-functional normalization and canonical quantity

Let $A_i = A_i^* = A_i^2$ and $B_j = B_j^* = B_j^2$, for $i, j \in \{1, 2, 3\}$, be the event projectors of binary measurements. We use

$$\begin{aligned} \mathcal{B} = & -A_2 - B_1 - 2B_2 + A_1B_1 + A_1B_2 + A_2B_1 + A_2B_2 \\ & - A_1B_3 + A_2B_3 - A_3B_1 + A_3B_2. \end{aligned} \quad (2.1)$$

Here A_iB_j means $A_i \otimes B_j$; the omitted tensor factor in a marginal is the identity. The convention uses event probabilities, not ± 1 observables.

Definition 2.1. The number S_t is the supremum of $\langle \psi, \mathcal{B}\psi \rangle$ over unit vectors $\psi \in \mathcal{H}_A \otimes \mathcal{H}_B$ and the six projectors above, where both local Hilbert spaces are finite-dimensional and their dimensions are not fixed in advance. Equivalently one may use normalized density operators: a finite spectral decomposition expresses their values as convex combinations of pure-state values for the same measurements.

Proposition 2.2 (Normalization identities). *The local bound of (2.1) is zero. In the event convention of Mghirbi [4], the substitution*

$$(P_1, P_2, P_3) = (A_2, A_1, A_3), \quad (Q_1, Q_2, Q_3) = (B_2, B_1, B_3)$$

identifies the Bell expression exactly, with scale one and shift zero. Writing $X_i = 2P_i - I$, $Y_j = 2Q_j - I$, its involution version is

$$\begin{aligned} 4\mathcal{B} &= \mathcal{B} - 4I, \\ \mathcal{B} &= X_1 + X_2 - Y_1 - Y_2 + X_1(Y_1 + Y_2 + Y_3) \\ &\quad + X_2(Y_1 + Y_2 - Y_3) + X_3(Y_1 - Y_2). \end{aligned} \quad (2.2)$$

Proof. Substitution in $I_M = -P_1 - 2Q_1 - Q_2 + P_1Q_1 + P_1Q_2 + P_1Q_3 + P_2Q_1 + P_2Q_2 - P_2Q_3 + P_3Q_1 - P_3Q_2$ gives every coefficient of (2.1). Expanding $P_i = (I + X_i)/2$, $Q_j = (I + Y_j)/2$ gives (2.2). These maps are invertible on the stated projective strategy class and preserve dimensions and the state. Exact evaluation on the 64 deterministic assignments of six bits gives maximum zero and minimum -3 ; all-zero projectors attain zero. Convexity gives the same local bound for mixtures. The coefficient identities and all 64 integer evaluations are recomputed in the normalization certificate of Appendix B. $\square$

We will use operators on $\ell^2(\mathbb{Z})$ as variational carriers. Their appearance does not redefine S_t . The equality linking their variational parameter to the finite tensor supremum is proved in Proposition 3.3; it uses finite realizations and a specifically restricted external upper-bound theorem.

3 Canonical upstream reduction

3.1 A certified parameter interval

Proposition 3.1 (Finite certificates for the initial interval). *With the exact rationals L, U of Theorem 1.1,*

$$L < S_t \leq U. \quad (3.1)$$

Proof. The lower certificate reconstructs six rational orthogonal projectors on $\mathbb{C}^{499}$ and a nonzero rational Schmidt vector. The exact circle identities for all 498 adjacent block parameters make the blocks projective. Dividing their direct Bell contraction by the positive squared norm gives a rational value v satisfying

$$0.2508753845139765354 \leq v < 0.2508753845139765355.$$

It is an actual finite tensor value, so $L < v \leq S_t$.

For the upper bound, let $\mathbf{w}$ be the column of 244 reduced words of total length at most four in the six involutions of (2.2). Words are reduced using adjacent-square cancellation within each party and commutation across parties only. The upper certificate constructs a rational positive semidefinite 244×244 matrix Y and checks the noncommutative polynomial identity

$$\frac{2501750771}{500000000}I - \mathbf{B} = \mathbf{w}^*Y\mathbf{w}. \quad (3.2)$$

The positivity check consists of 183 positive exact leading principal determinants of five integer blocks, followed by a positive congruence. Expansion of all 59,536 ordered word products has zero residual at all 8,452 coefficient positions. Evaluating the identity in any finite tensor strategy and any vector gives a nonnegative right-hand expectation. Equation (2.2) therefore gives

$$\mathcal{B} \leq \frac{1}{4} \left(\frac{2501750771}{500000000} - 4 \right) I = \frac{501750771}{2000000000} I = UI.$$

This constant is uniform in the finite dimensions, so it bounds their supremum. Appendix B specifies the rational inputs, block matrices, reconstruction and acceptance predicates. No numerical solver output, hierarchy convergence theorem or symmetry-completeness assertion is needed for this operator implication. $\square$

3.2 Precisely restricted external premises

Put

$$w(c) = \sqrt{1 - c^2}, \quad b(c) = \frac{w(c)}{2}, \quad d(x, y) = xy + \frac{x - y}{2} - 1. \quad (3.3)$$

For a finite path $\gamma = (\gamma_0, \dots, \gamma_m)$ with $\gamma_0, \gamma_m \in \{-1, 1\}$ and the remaining labels in $[-1, 1]$, define

$$(J_m(\gamma))_{ii} = d(\gamma_{i-1}, \gamma_i), \quad (J_m(\gamma))_{i,i+1} = b(\gamma_i), \quad q^* = \sup_{m, \gamma} \lambda_{\max}(J_m(\gamma)). \quad (3.4)$$

The matrices are real symmetric; unspecified entries vanish.

External premise 3.2 (Coladangelo v1). For the functional (2.1) and the number q^* in (3.4), we use exactly the following three source interfaces:

1. Theorem 7.1 of [2] (p. 24, (33)–(34): there is a finite, nonnegative, concave function $\mathfrak{h} : [-1, 1] \rightarrow [0, \infty)$ such that, on the closed intervals,

$$\mathfrak{h}(y) + d(x, y) + \mathfrak{h}(-x) \leq q^*, \quad \mathfrak{h}(z)\mathfrak{h}(-z) \geq b(z)^2. \quad (3.5)$$

2. The finite-dimensional projective tensor restriction of Theorem 7.2 of that source (p. 24): every normalized strategy in Definition 2.1 has value at most q^* .
3. Proposition 10.9 of that source (pp. 70–71): there are finite paths $1 = \gamma_0^{(n)} \geq \dots \geq \gamma_{m_n}^{(n)} = -1$ whose largest Jacobi eigenvalues tend to q^* .

These are three consumed interfaces from one source, with shared proof ancestry. In particular the upper-bound theorem uses the scalar-certificate theory. They are not three independent external confirmations. No global differentiability or strict concavity of $\mathfrak{h}$, and no classification of its entire equality graph, is part of the premise. The source's broader representation statements are not imported.

Proposition 3.3 (Finite realization and value identification). *Every largest eigenvalue in (3.4) is realized by a finite-dimensional projective tensor strategy for (2.1). Consequently $q^* = S_t$. Moreover for every bilateral sequence $c_j \in [-1, 1]$, the bounded self-adjoint Jacobi operator*

$$(H(c)f)_j = b(c_j)f_{j-1} + d(c_j, c_{j+1})f_j + b(c_{j+1})f_{j+1} \quad (3.6)$$

satisfies $H(c) \leq S_t I$.

Proof. Appendix A.1 constructs the six projectors explicitly by alternating two-dimensional blocks. On $\sum_i v_i e_i \otimes e_i$, the Bell expectation is exactly $v^T J_m(\gamma)v$. A real unit top eigenvector therefore proves $q^* \leq S_t$. External premise 3.2(2) proves the reverse inequality in precisely the same finite projective tensor class.

On $[-1, 1]^2$, $-3 \leq d \leq 0$, since its four corner values are $0, -1, -3, 0$ and bilinear interpolation has nonnegative weights. Also $0 \leq b \leq 1/2$, hence $\|H(c)\| \leq 4$. For any finitely supported real f , retain all labels incident on its support, add a zero-amplitude vertex at either end, and choose the two new outer labels in $\{-1, 1\}$. The resulting finite path has the same Rayleigh numerator. Its value is at most $q^* \|f\|^2$. Density and boundedness extend the inequality to $\ell^2(\mathbb{Z})$; real and imaginary parts give the complex version. This comparison permits arbitrary finite label changes, with no monotonicity constraint. $\square$

4 Outer stationary carriers and recurrence

For $s \in I$, define

$$a(s) = \sqrt{\frac{4s + 5 + \sqrt{16s^2 + 24s - 7}}{8}}, \quad (4.1)$$

$$r(c) = \frac{\sqrt{1 - c^2}(2c - 1)}{(1 - c)(2c + 1)}. \quad (4.2)$$

The ratio formula is used at the outer labels $\pm a(s)$, where all its denominators are nonzero. The stationary map on state coordinates (x, y, u) is

$$v = \frac{2(s - d(x, y)) - w(x)/u}{w(y)}, \quad z = \frac{(1 - 2x) + 2yv/w(y)}{2v^2} - \frac{1}{2}, \quad (4.3)$$

$$F_s(x, y, u) = (y, z, v).$$

Its underlying state domain is $\mathcal{D} = (-1, 1)^2 \times (0, \infty)$. A *valid step* requires that (4.3) be defined and that $z \in (-1, 1)$, $v > 0$. Every iterate used in this paper includes validity of all intervening steps. A chain is decreasing if its states are (c_j, c_{j+1}, u_j) with $c_j > c_{j+1}$, and all its steps are valid.

Definition 4.1 (Admissible carriers). For $s \in I$, $\mathcal{C}_s$ consists of real bilateral pairs (c, λ) satisfying

$$\begin{aligned} -1 < c_j < 1, \quad c_j > c_{j+1}, \quad \lambda_j > 0, \quad \sum_{j \in \mathbb{Z}} \lambda_j^2 = 1, \quad H(c)\lambda = s\lambda, \\ u_j = \lambda_j / \lambda_{j-1}, \quad F_s(c_j, c_{j+1}, u_j) = (c_{j+1}, c_{j+2}, u_{j+1}) \quad \text{by valid steps}, \\ (c_j, u_j) \longrightarrow (a(s), r(a(s))) \quad (j \rightarrow -\infty), \\ (c_j, u_j) \longrightarrow (-a(s), r(-a(s))) \quad (j \rightarrow +\infty). \end{aligned} \quad (4.4)$$

These are admissibility conditions. Membership at a parameter s does not itself identify s with the Bell supremum.

Theorem 4.2 (Existence of an outer carrier at the supremum). *Under External premise 3.2 and Proposition 3.1, $\mathcal{C}_{S_t} \neq \emptyset$. At its tails $r(a(S_t)) > 1$ and $0 < r(-a(S_t)) < 1$.*

Proof. We give the construction and the necessary tail selection in Appendix A, with the following logical order. For the monotone finite approximants supplied by External premise 3.2(3), the variation estimate

$$\langle f, J_m(\gamma)f \rangle \leq \frac{1}{4} \|f\|_2^2 + \frac{1}{2} \text{TV}(\gamma) \|f\|_\infty^2$$

prevents unit top eigenvectors from vanishing after a maximum coordinate is translated to zero. A coordinatewise limit gives a nonzero nonnegative ℓ^2 eigenvector at $S_t = q^*$ and nonincreasing labels. The universal Jacobi comparison in Proposition 3.3 excludes finite and half-line support by an explicit endpoint perturbation. All labels are therefore interior and all amplitudes positive.

Finite label variations give stationarity and (4.3). Equality in the two scalar inequalities (3.5) gives differentiability of the storage function at saturated labels and uniqueness of their saturated predecessor and successor. A plateau would propagate to a constant carrier, whose value is at most $1/4$. Thus labels decrease strictly.

The remaining steps cannot be replaced by assuming an outer tail. Endpoint limits ± 1 are excluded directly from stationarity and the eigenvalue equation. A positive Jacobi ratio lemma proves actual ratio limits and uses square summability to select their sides of one. Eliminating these limits gives

$$(s - t^2)\{4t^4 - (4s + 5)t^2 + s + 2\} = 0.$$

A five-site second variation excludes $s = t^2$. The quartic has inner and outer magnitudes. A concavity argument excludes mixed magnitudes, and strict decrease excludes the remaining inner pair. This yields exactly the four outer limits in (4.4), not an assumption of symmetry of the full carrier. $\square$

For clarity, the recurrence itself follows from two different equations. At site j , divide the Jacobi equation by λ_j . At label $j + 1$, differentiate the finite label variation of the Rayleigh form and divide by λ_j^2 . With $(x, y, z) = (c_j, c_{j+1}, c_{j+2})$ and $(u, v) = (\lambda_j / \lambda_{j-1}, \lambda_{j+1} / \lambda_j)$, the results are

$$w(x)/u + w(y)v = 2(s - d(x, y)), \quad (x - 1/2) + (z + 1/2)v^2 - yv/w(y) = 0. \quad (4.5)$$

Solving these equations is precisely (4.3). Positivity of the amplitudes and interior labels justify all divisions.

Theorem 4.2 is existential. It neither classifies every optimizer nor asserts carrier uniqueness, finite-dimensional attainment, or a symmetry of all optimal strategies. This limited conclusion is sufficient: the subsequent uniqueness theorem concerns the matching *parameter*.

5 Local dynamics and full-state matching

5.1 Outer fixed points and hyperbolicity

Use the outer coordinate

$$t = a(s), \quad \sigma(t) = \frac{4t^4 - 5t^2 + 2}{4t^2 - 1}, \quad T = a(I) = (a(L), a(U)). \quad (5.1)$$

The equation $s = \sigma(t)$ is $4t^4 - (4s + 5)t^2 + s + 2 = 0$; its larger squared root is (4.1). On this branch $\sqrt{3}/2 < t < 1$, and

$$\sigma'(t) = \frac{2t(4t^2 - 3)(4t^2 + 1)}{(4t^2 - 1)^2} > 0. \quad (5.2)$$

Exact comparison through σ gives $0.8782729451808 < a(s) < 0.87827295218$ on all of I , including its upper endpoint. This is an enclosure of the full initial domain, not a restriction to a neighborhood of a suspected root.

External premise 5.1 (Real-analytic hyperbolic local manifolds). A real-analytic local diffeomorphism at a hyperbolic fixed point has analytic local stable and unstable manifolds tangent to the corresponding invariant subspaces. In sufficiently small admissible neighborhoods their local convergence sets capture the appropriate forward and backward convergent orbits; the manifold germs are unique. We use the standard finite-dimensional local theorem, with analytic regularity as in [1], Theorem 1.1 and the stable/unstable specialization, and the usual local capture formulation [7].

Lemma 5.2 (The intrinsic local tails). *For every $s \in I$,*

$$p_{\pm}(s) = (\pm a(s), \pm a(s), r(\pm a(s)))$$

are valid fixed points. The positive fixed point has one unstable and two stable directions; the negative fixed point has one stable and two unstable directions. Their splittings are uniform over the closed parameter hull. The decreasing backward convergence set near p_+ and decreasing forward convergence set near p_- are the corresponding one-dimensional analytic local tail pieces.

Proof. Substitution in (4.3) proves the fixed-point equations and $r(t)r(-t) = 1$. Further,

$$r(t)^2 - 1 = \frac{2t(4t^2 - 3)}{(1 - t)(2t + 1)^2} > 0.$$

At p_+ , one eigenvalue is $\nu = 1/r(t)^2 \in (0, 1)$. The other two are the zeros of the sign-equivalent characteristic factor

$$(4t^3 - 3t + 1)\xi^2 - 4t^2(4t^2 - 1)\xi - (4t^3 - 3t - 1).$$

Its leading and constant coefficients are positive, its root sum is positive, and its value at one is $-2(2t^2 - 1)(4t^2 + 1) < 0$. Hence its roots satisfy $0 < \alpha < 1 < \mu$. The spectrum at p_- is reciprocal. The determinants at the two points are respectively

$$\frac{(t - 1)^2(2t + 1)^4}{(t + 1)^2(2t - 1)^4}, \quad \frac{(t + 1)^2(2t - 1)^4}{(t - 1)^2(2t + 1)^4},$$

both nonzero. These are exact symbolic identities, specified in Appendix C. Positive radicands and ratios make F_s analytic near the points; its nonzero determinant gives local invertibility. Compactness of $[a(L), a(U)]$ and continuous eigenvalue branches, none on the unit circle, give uniform hyperbolicity. External premise 5.1 now supplies the stated manifolds and local capture. $\square$

5.2 Intrinsic definitions and the crossing convention

Choose sufficiently small admissible open neighborhoods $N_\pm$ of the fixed points. Define $W_{s,\text{loc}}^u$ to consist of states in N_+ admitting a valid backward chain wholly in N_+ which converges to p_+ ; define $W_{s,\text{loc}}^s$ using a valid forward chain wholly in N_- converging to p_- . Superscript $\downarrow$ restricts these sets to decreasing tails. In particular it excludes the fixed point itself. These two tail sets are defined independently.

Definition 5.3 (Full-state matching). For $s \in I$, define

$$\begin{aligned}\mathcal{U}_s &= \{X : F_s^m(\xi) = X \text{ along a valid decreasing chain,} \\ &\quad m \geq 0, \xi \in W_{s,\text{loc}}^{u,\downarrow}\}, \\ \mathcal{V}_s &= \{X : F_s^n(X) = \eta \text{ along a valid decreasing chain,} \\ &\quad n \geq 0, \eta \in W_{s,\text{loc}}^{s,\downarrow}\}.\end{aligned}$$

The integers and the two local tail points are independent. Set

$$\Sigma = \{(x, y, u) \in \mathcal{D} : x \geq 0 > y\}, \quad \mathcal{M}_s = \Sigma \cap \mathcal{U}_s \cap \mathcal{V}_s, \quad (5.3)$$

and $\mathcal{M} = \{(s, X) : s \in I, X \in \mathcal{M}_s\}$. All three state coordinates must agree at the join. Equality of the two labels alone is insufficient. The half-open crossing set chooses a representative of a discrete orbit; smooth transversality is not assumed.

The global continuations are independent of the sufficiently small admissible neighborhoods. Indeed a backward tail tending to p_+ eventually lies entirely in any second such neighborhood. An earlier point on that same tail represents the same global state by finitely many valid decreasing steps. Interchanging the neighborhoods gives equality of the source continuations; the forward argument gives equality of the target continuations.

Proposition 5.4 (Matching soundness and carrier completeness). *For every $s \in I$,*

$$\mathcal{C}_s \neq \emptyset \iff \mathcal{M}_s \neq \emptyset. \quad (5.4)$$

Every carrier has a unique index k with $c_k \geq 0 > c_{k+1}$, and its state there belongs to $\mathcal{M}_s$. Every match reconstructs an admissible normalized positive carrier.

Proof. Given a match, concatenate its backward and forward tails at their identical three-coordinate state. This produces a bilateral valid decreasing chain (c_j, c_{j+1}, u_j) with the four prescribed limits. Choose $\lambda_0 > 0$ and impose $\lambda_j = u_j \lambda_{j-1}$ in both directions. All amplitudes are positive. On the far left, ratios are bounded below by some $q_- > 1$, so amplitudes decay geometrically when moving left. On the far right they are bounded above by some $q_+ < 1$, so amplitudes decay geometrically when moving right. Thus $0 < \sum \lambda_j^2 < \infty$, and a single scaling normalizes the vector. The first equation in (4.5), multiplied by $\lambda_j/2$, gives exactly $H(c)\lambda = s\lambda$. The second equation remains true at every chain step. All conditions of Definition 4.1 follow.

Conversely, let $(c, \lambda) \in \mathcal{C}_s$. Strict decrease and the positive left and negative right limits imply that $\{j : c_j \geq 0\}$ is a nonempty initial segment of $\mathbb{Z}$ bounded above. Its maximum k is the unique crossing index. If $c_k = 0$, the representative starts at that zero and ends at the next negative label, as required by (5.3). The full states $X_j = (c_j, c_{j+1}, \lambda_j/\lambda_{j-1})$ tend to p_+ and p_- . Hence for some finite $m, n \geq 0$, the entire past of X_{k-m} lies in N_+ and the entire future of X_{k+n} lies in N_- . Local capture in Lemma 5.2 places these states on the decreasing unstable and stable pieces. Their actual carrier steps give $F_s^m(X_{k-m}) = X_k$ and $F_s^n(X_k) = X_{k+n}$, with all validity conditions retained. Thus $X_k \in \mathcal{M}_s$. $\square$

Proposition 5.5 (A match at the true value). *The matching set $\mathcal{M}_{S_t}$ is nonempty.*

Proof. Proposition 3.1 puts S_t in the domain I . Theorem 4.2 supplies a member of $\mathcal{C}_{S_t}$. Apply Proposition 5.4 at that parameter. $\square$

6 Global uniqueness of the matching parameter

6.1 Rational coordinates and reversibility

The phase variables in this section are ζ, η . They are independent and must not be confused with the amplitude ratio u in F_s . Write $B_0(x) = b(x)^2 = (1 - x^2)/4$. On a valid complete chain set

$$\Phi_s(x, y, u) = (x, y, q), \quad q = s - d(x, y) - b(x)/u, \quad \bar{q} = s - d(x, y) - q. \quad (6.1)$$

Here $q = b(y)v > 0$, $\bar{q} = b(x)/u > 0$, and the inverse is $\Phi_s^{-1}(x, y, q) = (x, y, b(x)/\bar{q})$. Thus the relevant rational domain is $|x|, |y| < 1$, $q, \bar{q} > 0$. We do not assert that every state in the larger domain $\mathcal{D}$ has positive outgoing q .

Lemma 6.1 (Exact conjugacy, reverser and contact identities). *In these coordinates the propagation map and its inverse are*

$$\begin{aligned} \widehat{F}_s(x, y, q) &= (y, z, Q), \quad z = \frac{y}{2q} - \frac{(x - 1/2)B_0(y)}{q^2} - \frac{1}{2}, \\ Q &= s - d(y, z) - \frac{B_0(y)}{q}, \\ \widehat{F}_s^{-1}(x, y, q) &= (w_-, x, B_0(x)/\bar{q}), \\ w_- &= \frac{x}{2\bar{q}} - \frac{(y + 1/2)B_0(x)}{\bar{q}^2} + \frac{1}{2}. \end{aligned} \quad (6.2)$$

The involution

$$R_s(x, y, q) = (-y, -x, \bar{q}) \quad (6.3)$$

satisfies $R_s \widehat{F}_s R_s = \widehat{F}_s^{-1}$, exchanges $q, \bar{q}$, and preserves decreasing label orientation under reversal of chains. For fixed s , the contact form $\omega = dq + (x - 1/2) dy$ satisfies

$$\widehat{F}_s^* \omega = \frac{B_0(y)}{q^2} \omega, \quad R_s^* \omega = -\omega, \quad \det D\widehat{F}_s = \left(\frac{B_0(y)}{q^2} \right)^2. \quad (6.4)$$

Proof. Substitute $v = q/b(y)$ in (4.3). At its image the complementary coordinate is $B_0(y)/q$; solving the resulting equations gives both inverse identities. The equality $d(-y, -x) = d(x, y)$ then proves (6.3). For the contact calculation,

$$dQ + (y - 1/2) dz = \left(-(z + 1/2) + \frac{y}{2q} \right) dy + \frac{B_0(y)}{q^2} dq;$$

the formula for z makes its dy coefficient $(x - 1/2)B_0(y)/q^2$. Differentiation also gives the stated determinant. Appendix C specifies the exact polynomial-cancellation checks, including comparison with the implemented map and Jacobian. $\square$

Reversal takes the intrinsic positive unstable tail to the intrinsic negative stable tail. This describes the second tail as a set; it does not assume that a given orbit meets its own reflection.

6.2 An exactly defined analytic tail

Put

$$G(t) = \frac{(2t-1)(t+1)}{2(2t+1)}, \quad p_t = (t, t, G(t)), \quad (6.5)$$

$$\begin{aligned} A(t) &= (t+1)(2t-1)^2, & C(t) &= (1-t)(2t+1)^2, & D(t) &= 4t^2(4t^2-1), \\ \mu(t) &= \frac{D(t) + \sqrt{D(t)^2 - 4A(t)C(t)}}{2A(t)}, & \rho(t) &= \frac{C(t)}{A(t)}. \end{aligned} \quad (6.6)$$

At p_t , the characteristic polynomial is

$$(\xi - \rho)(\xi^2 - (D/A)\xi + \rho).$$

Thus its eigenvalues are $\mu, \rho, \rho/\mu$, with $\mu > 1$ and $0 < \rho, \rho/\mu < 1$ on T . Direct cancellation gives $G(t)/b(t) = r(t)$, so this is the same positive outer fixed point.

Define $P(t, \zeta)$ as the analytic invariant germ specified by

$$\begin{aligned} P(t, 0) &= p_t, & P_\zeta(t, 0) &= (-1/\mu, -1, t - 1/2), \\ P(t, \mu\zeta) &= \widehat{F}_{\sigma(t)}(P(t, \zeta)). \end{aligned} \quad (6.7)$$

The normalization fixes the phase scale; it is part of the exact definition of the scalar residual, not a fitted numerical convention.

Finite certificate 6.2 (Analytic-germ enclosure). *Let $T^\square$ be the closed outward interval specified in Appendix C.2, which contains $[a(L), a(U)]$. On a complex parameter rectangle of half-width 10^{-5} about its midpoint and on the phase disk of radius $r_0 = 0.01$, the degree-36 coefficient recursion for (6.7) has a unique analytic correction. The degree-48 recursion encloses the same germ. With the sup norm on its three coordinates, the corrections satisfy respectively*

$$\|P - P_{36}\| < 3.414 \times 10^{-47}, \quad \|P - P_{48}\| < 8.334 \times 10^{-63}.$$

The parameter margin exceeds 9.996×10^{-6} ; division denominators are separated from zero. The contraction factors are less than 6.891×10^{-27} and 4.409×10^{-36} , respectively. The exact inequalities, coefficient boxes and margins are the outputs of the local certificates specified in Appendix C.2.

Analytic implication of the certificate. Write $P_N = \sum_{n=0}^N c_n(t)\zeta^n$. Once c_0, c_1 are fixed by (6.7), set the trial coefficient c_n to zero and compute $b_n = [\zeta^n]\widehat{F}_{\sigma(t)}(P_{<n})$. Then

$$(\mu^n \text{Id} - J)c_n = b_n, \quad J = D\widehat{F}_{\sigma(t)}(p_t), \quad n \geq 2.$$

The spectral separation makes these equations unique; interval Cramer's rule additionally rejects a determinant enclosure containing zero. This recursion is regenerated over the complex rectangle, with the positive-real-part square-root branch explicitly certified.

Let $R_0 = 0.2$ and $D_N(\zeta) = \widehat{F}_{\sigma(t)}(P_N(\zeta/\mu)) - P_N(\zeta)$. The first $N+1$ coefficients of D_N vanish. A coefficient majorant bounds its norm by M on $|\zeta| \leq R_0$; hence on the smaller disk

$$\|D_N\| \leq \delta_N := M \frac{(r_0/R_0)^{N+1}}{1 - r_0/R_0}.$$

For analytic corrections vanishing to order $N + 1$, Schwarz's estimate gives $\|e(\zeta/\mu)\| \leq \|e\|/m^{N+1}$, where the certified lower bound is $m > 5.8365$. An enclosed Jacobian row-sum norm L_F gives the contraction factor $\kappa = L_F/m^{N+1}$ for the correction map

$$(\mathcal{T}_N e)(t, \zeta) = \widehat{F}_{\sigma(t)}(P_N(t, \zeta/\mu(t)) + e(t, \zeta/\mu(t))) - P_N(t, \zeta).$$

It preserves the required vanishing order by the coefficient equations. The finite gates are $\kappa < 1/2$ and $\delta_N + \kappa\epsilon_N \leq \epsilon_N$. They make the correction map a self-map and contraction on the ball of radius ϵ_N . Banach's theorem yields the exact analytic germ; conjugation and uniqueness give reality on real arguments.

Cauchy estimates give parameter and phase derivative errors bounded by ϵ_N/Δ_t and $\epsilon_N/(r_0 - |\zeta|)$. These errors are added before chart propagation. All subsequent interval statements therefore concern P , not only its polynomial truncation. The germ is injective since $y_\zeta(t, 0) = -1$, and backward iteration divides its phase by μ . Its tangent and backward convergence, together with Lemma 5.2, identify it with the same intrinsic unstable manifold. Its negative phase side has $x - y < 0$ near zero and cannot represent a decreasing outer tail. The positive side does. Finally the two degrees have identical uniquely determined Taylor coefficients at every order; their analytic germs and valid continuations coincide. $\square$

The contact relation has a useful exact consequence. For $f(\zeta) = \omega(P_\zeta(\zeta))$, invariance gives

$$f(\zeta) = \frac{B_0(y(\zeta/\mu))}{q(\zeta/\mu)^2 \mu} f(\zeta/\mu).$$

The multiplier at zero is $\rho/\mu < 1$. Iteration and boundedness imply $f = 0$ on a small germ, and analytic continuation through valid iterates gives

$$q_\zeta = (1/2 - x)y_\zeta. \quad (6.8)$$

This identity is proved analytically, rather than inferred from a small numerical defect.

6.3 Completeness of the two-phase domain

Define the propagated chart

$$W(t, \zeta) = \widehat{F}_{\sigma(t)}^5 P(t, \zeta) = (x(t, \zeta), y(t, \zeta), q(t, \zeta)), \quad \zeta_a = 0.000078125, \quad \zeta_b = 0.0025. \quad (6.9)$$

The phase interval is generated from $r_0/128, r_0/4$, and the iterate count is the first nonnegative count for which the right endpoint has strictly negative x . The frozen computation gives five.

Finite certificate 6.3 (Complete chart geometry). *For all $t \in T^\square$, $\zeta \in [0, \zeta_b]$, and all iterates $0, \dots, 5$, the analytic chart obeys $|x|, |y| < 1$, $q, \bar{q} > 0$, and $x_\zeta, y_\zeta < 0$. At the two envelope endpoints,*

$$y(t, \zeta_a) > 0, \quad x(t, \zeta_b) < 0, \quad x(t, \zeta_a) + y(t, \zeta_b) > 0.$$

The geometry certificate covers the closed phase interval with 14 leaves. Every leaf contains enclosures for each intermediate iterate; its parameter domain is the whole $T^\square$.

The certificate is a finite interval statement about the corrected chart of Certificate 6.2. Its verification checks the signs of interval endpoints and exact joins in the subdivision tree (Appendix C.2).

Lemma 6.4 (All intrinsic matches enter the phase envelope). *Every match in $\mathcal{M}_s$, after Φ_s , satisfies*

$$W(t, \zeta) = R_{\sigma(t)} W(t, \eta), \quad t = a(s), \quad (\zeta, \eta) \in [\zeta_a, \zeta_b]^2. \quad (6.10)$$

The phases are independent. This closed product covers the canonical exact-zero crossing representatives.

Proof. The functional equation gives

$$\widehat{F}_{\sigma(t)} W(t, \zeta/\mu) = W(t, \zeta), \quad x(t, \zeta) = y(t, \zeta/\mu).$$

Since y decreases strictly and $\mu > 1$, $x > y$ for every positive phase. The same identity applies to each earlier iterate. Certificate 6.3 thus proves validity and decreasing orientation on the entire positive propagated chart.

There are unique zeros $\zeta_y, \zeta_x \in (\zeta_a, \zeta_b)$ of y, x , and scaling gives $\zeta_x = \mu\zeta_y$. The precise source crossing domain is $(\zeta_y, \zeta_x]$. Local capture identifies any intrinsic decreasing outer tail with a positive phase of P , which can be made arbitrarily small by moving backward. Successive steps multiply the phase by μ . A first step past ζ_b cannot occur before crossing: its preceding phase would exceed $\zeta_b/\mu > \zeta_y$, where y is already negative. Thus the first crossing lies in $(\zeta_y, \mu\zeta_y]$, with every preceding iterate certified. Strict decrease precludes a second crossing. This also rules out an excursion outside the certified chart before the representative.

Apply the same reasoning independently to the reversed stable tail. Reversal exchanges the positive $q, \bar{q}$ coordinates and preserves validity and decreasing orientation. If the canonical representative has $x = 0 > y$, its reflection has positive first label and second label zero. Accordingly the reflected target domain is $[\zeta_y, \zeta_x)$, whereas the source domain is $(\zeta_y, \zeta_x]$. Both lie in the closed product used in (6.10). No exact-zero state is lost. $\square$

6.4 Projected inversion and exclusion of asymmetric full matches

The first label equation in (6.10) is

$$x(t, \eta) + y(t, \zeta) = 0. \quad (6.11)$$

For a possible source crossing, $x(t, \zeta) \geq 0$ and $y(t, \zeta) \leq 0$. The endpoint and monotonicity gates give

$$x(t, \zeta_a) + y(t, \zeta) > 0, \quad x(t, \zeta_b) + y(t, \zeta) < 0, \quad x_\eta < 0.$$

The intermediate value theorem and strict monotonicity therefore give one and only one target phase $\eta = \eta(t, \zeta)$ in the entire envelope. In particular, eliminating this phase does not assume a branch near a candidate point. The inverse is analytic wherever used, and

$$\eta_\zeta = -\frac{y_\zeta(t, \zeta)}{x_\eta(t, \eta)}, \quad \eta_t = -\frac{y_t(t, \zeta) + x_t(t, \eta)}{x_\eta(t, \eta)}. \quad (6.12)$$

On this projected graph, define the two remaining residuals

$$\begin{aligned} \mathcal{H}(t, \zeta) &= x(t, \zeta) + y(t, \eta(t, \zeta)), \\ \mathcal{K}(t, \zeta) &= q(t, \zeta) + q(t, \eta(t, \zeta)) - \sigma(t) + d(x(t, \zeta), y(t, \zeta)). \end{aligned} \quad (6.13)$$

Projection together with $\mathcal{H} = \mathcal{K} = 0$ is equivalent to the complete three-coordinate equation: if $\mathcal{H} = 0$, both label pairs are reflected and $d(x(t, \eta), y(t, \eta)) = d(x(t, \zeta), y(t, \zeta))$; then $\mathcal{K} = 0$ is exactly the complementary- q equation. Neither label equation alone checks the amplitude ratio.

The contact identity (6.8) and the projected equation, without assuming $\mathcal{H} = 0$, give

$$\mathcal{H}_\zeta = x_\zeta(t, \zeta) - \frac{y_\eta(t, \eta)y_\zeta(t, \zeta)}{x_\eta(t, \eta)}, \quad \mathcal{H}_\zeta = (y(t, \zeta) + 1/2)\mathcal{H}_\zeta. \quad (6.14)$$

These exact identities control interval overestimation on projected cells. Parameter derivatives instead use the full chain rule (6.12); the fixed-parameter contact relation is not applied to differentiation in t .

Finite certificate 6.5 (Exhaustive full-match exclusions). *Let $C_0 = [0.0005, 0.0007]$. Over the entire parameter box $T^\square$ and all independent target phases in $[\zeta_a, \zeta_b]$, a closed subdivision of the source envelope has 179 leaves. Each leaf passes one of these predicates:*

1. $x < 0$ or $y > 0$ throughout the cell (7 leaves);
2. the enclosure of $\mathcal{H}$ excludes zero (148 leaves);
3. the enclosure of $\mathcal{K}$ excludes zero (22 leaves);
4. the source cell lies in C_0 and $\mathcal{H}_\zeta > 0$ throughout its projected family (2 leaves).

A separate closed cover of all of C_0 , with 554 leaves, proves $\mathcal{H}_\zeta > 0$ there. On C_0 , $g = x + y$ has $g_\zeta < 0$, with $g(t, 0.0005) > 0$ and $g(t, 0.0007) < 0$. There are no unresolved leaves in either tree.

The finite input to each predicate is the analytic chart enclosure, not sampled points. The inverse enclosure starts from the entire target envelope and retains every solution under interval Newton intersection:

$$V_{\text{new}} = V \cap \left(m - \frac{x(t, m) + y(t, \zeta)}{[x_\eta](t, V)} \right).$$

The first derivative interval covers the full envelope. A floating-point scout selects only a center, which is checked within the envelope; it does not supply an exclusion or a target-phase restriction. Each retained family has certified strict endpoint signs for the inverse. Subsequent derivative intersections use already justified bounds. Tree acceptance requires the children to have precisely the parent's endpoints and a common join, every child to be reached exactly once, and every terminal predicate to be resolved. These specifications are detailed in Appendix C.2.

Proposition 6.6 (Diagonal phases are a conclusion). *For each $t \in T$ there is a unique analytic phase $\phi(t) \in C_0$ with $x(t, \phi) + y(t, \phi) = 0$. Every intrinsic full match has $\zeta = \eta = \phi(t)$.*

Proof. The signs and nonzero phase derivative of g give its unique root in C_0 , and the analytic implicit function theorem gives ϕ . At that root $\eta = \phi$ solves (6.11); uniqueness of projected inversion identifies it as the target phase. Hence $\mathcal{H}(t, \phi) = 0$. The separate full central cover proves strict monotonicity of $\mathcal{H}$ throughout C_0 , making ϕ its only zero there. The full cover excludes every noncentral full match by one of the first three predicates. Lemma 6.4 makes that cover exhaustive for the intrinsic matching set, including both zero boundaries. Thus every full match has the stated equal phases. $\square$

Asymmetric solutions of the label equations are not discarded by symmetry. The $\mathcal{H}$ -exclusion cells rigorously exclude the remaining off-central label branches by their third-coordinate mismatch.

6.5 Scalar equivalence and full-interval monotonicity

We can now define

$$\mathcal{R}(t) = 2q(t, \phi(t)) - \sigma(t) + d(x(t, \phi(t)), y(t, \phi(t))). \quad (6.15)$$

Equations (6.7), (6.9), the unique zero defining ϕ , and (6.15) are an exact structural definition. Truncated polynomials serve only to enclose this analytic function.

Proposition 6.7 (Scalar/full-match equivalence). *For every $t \in T$,*

$$\mathcal{R}(t) = 0 \iff \mathcal{M}_{\sigma(t)} \neq \emptyset.$$

Proof. Every full match is diagonal by Proposition 6.6; its third-coordinate equation is precisely $\mathcal{R} = 0$. Conversely, $\mathcal{R}(t) = 0$ and $g(t, \phi) = 0$ give $x = -y$ and $2q = \sigma - d(x, y)$. Thus $W(t, \phi) = R_{\sigma(t)}W(t, \phi)$ in all three coordinates. The positive-phase source tail and each of its five propagated steps are valid and decreasing. Reflecting this chain gives a valid decreasing forward tail to the negative fixed point. Since $x > y$ and $x = -y$, the joined state has $x > 0 > y$. Applying $\Phi_{\sigma(t)}^{-1}$ recovers every amplitude ratio and places the join in the canonical Σ , $\mathcal{U}_s$ and $\mathcal{V}_s$. This is a genuine member of $\mathcal{M}_s$. $\square$

Implicit differentiation gives $\phi' = -(x_t + y_t)/(x_\zeta + y_\zeta)$. Using (6.8) in the full derivative of (6.15) cancels the phase terms and yields

$$\mathcal{R}'(t) = 2q_t - \sigma'(t) + (2x - 1)y_t \quad \text{at } (t, \phi(t)). \quad (6.16)$$

The partial derivatives on the right hold the phase fixed.

Finite certificate 6.8 (Global derivative and endpoint signs). *On the entire closed box $T^\square$, the corrected analytic chart and the interval enclosure of the whole phase family give*

$$3.4800 < \mathcal{R}'(t) < 3.5707.$$

The analytic extension through $a(L)$ satisfies

$$\mathcal{R}(a(L)) \in [-8.905044, -8.905043] \times 10^{-16}, \quad \mathcal{R}(a(U)) \in [2.4652543, 2.4652544] \times 10^{-8}.$$

These displayed outward bounds contain the full exact decimal intervals in `scalar.json`; they are checked as rational comparisons.

This certificate encloses $\phi(T^\square)$ as an interval family and evaluates (6.16) on the full parameter interval. It is not a set of derivative samples. Appendix C.2 gives the full derivative receipt and its finite acceptance rule.

Theorem 6.9 (Global matching-parameter uniqueness). *The projection*

$$\pi_s(\mathcal{M}) = \{s \in I : \mathcal{M}_s \neq \emptyset\}$$

is a singleton $\{s_\}$.*

Proof. The strict derivative makes $\mathcal{R}$ increasing on $T^\square$, so it has at most one zero there. Proposition 6.7 and the strict increase of σ then show that any two matching parameters in I are equal. The strict endpoint signs and continuity give a zero strictly between $a(L)$ and $a(U)$. The analytic extension through the excluded left endpoint justifies this use of the intermediate value theorem. Scalar equivalence supplies an intrinsic full match at that zero. This existence calculation is independent of using Proposition 5.5 as an existence premise, and uses no fine root enclosure as an input. $\square$

7 Exact-value identification

Theorem 7.1 (Identification of the canonical supremum). *The finite-dimensional projective tensor supremum is the unique matching parameter: $S_t = s_*$.*

Proof. The inference uses precisely the same interval, normalization, recurrence, ratio coordinate and intrinsic matching objects throughout:

$$\begin{array}{ll} S_t \in I & \text{Proposition 3.1,} \\ \mathcal{C}_{S_t} \neq \emptyset & \text{Theorem 4.2,} \\ \mathcal{M}_{S_t} \neq \emptyset & \text{Proposition 5.4 at } s = S_t, \\ S_t \in \pi_s(\mathcal{M}) & \text{definition of the projection,} \\ \pi_s(\mathcal{M}) = \{s_*\} & \text{Theorem 6.9.} \end{array}$$

Membership in the singleton gives $S_t = s_*$. Proposition 6.7 gives the equivalent residual characterization. $\square$

This composition uses one carrier at the true value and uniqueness of the parameter. It does not require uniqueness of a carrier or optimizer, a finite-dimensional attaining strategy, any assertion labelled R2, or commuting=tensor equality.

8 Certified numerical enclosure

Theorem 8.1 (Strict rational bracket). *For the rational numbers ℓ, h displayed after Theorem 1.1,*

$$\ell < S_t < h,$$

with the exact width in (1.1).

Proof. The degree-48 analytic enclosure of Certificate 6.2 represents the same germ as the degree-36 global proof. It is used only after the global equivalence and uniqueness theorems. Put $f(s) = \mathcal{R}(a(s))$; then

$$f'(s) = \frac{\mathcal{R}'(a(s))}{\sigma'(a(s))} > 0.$$

The exact directed interval for this quotient is recorded in `scalar.json`. Starting at the full closed interval $[L, U]$, the certificate performs 13 inclusions

$$J_{k+1} \supseteq J_k \cap \left(m_k - \frac{[f](m_k)}{[f']([L, U])} \right), \quad m_k \in J_k. \quad (8.1)$$

Every recorded output lies inside its input and strictly shrinks it. The mean value theorem shows that the root remains in each exact intersection; rational verification checks that the stored rounded output contains that intersection. The last output is inflated by exactly 10^{-52} on each side to give $[\ell, h]$.

The endpoint residual enclosures, generated for f at these s -valued endpoints, have the following outward rational summaries (the full decimal intervals are in `root.json`):

$$\begin{aligned} [f](\ell) &\subseteq [-5.720120, -5.720084] \times 10^{-51}, \\ [f](h) &\subseteq [5.804107, 5.804143] \times 10^{-51}. \end{aligned}$$

Their strict signs and continuity place a zero strictly between ℓ, h . Theorem 6.9 and scalar equivalence identify it with the unique global matching parameter; Theorem 7.1 identifies that parameter with S_t . Exact integer subtraction proves (1.1), and rational comparisons prove $L < \ell < h < U$. $\square$

The finite check of (8.1) uses integer/Fraction arithmetic, independently of the interval code’s midpoint helper: each recorded center is verified as an exact rational point inside its own input. The residual and derivative enclosures still depend on the analytic-germ proof and the interval arithmetic. Checking the final rational arithmetic alone would not establish those analytic enclosures.

Scouting suggested computational centers and a central phase interval. Proof acceptance instead uses rational identities, directed interval bounds, full covers and strict signs. Historical root digits, stored success flags and agreement of floating-point optimizers supply no premise of this enclosure.

9 Reproducibility and machine certificates

The manuscript is assembled from one authenticated input archive. Its primary mathematical authority is the final clean closure, which contains the canonical upstream and local-matching components and the surviving global-uniqueness component. The two other top-level archives are validation and source-provenance evidence. Their audit verdicts support confidence; the mathematical premises are the statements and proofs reproduced here, the cited external theorems, and the specified finite certificates.

The accompanying manuscript archive contains the original input ZIP, unaltered, under `reproducibility/`. The standard-library wrapper `tools/replay_frozen.py` offers the command

```
python tools/replay_frozen.py -full
```

from the manuscript directory. It checks the manuscript inventory, the original archive hash and its input manifest, extracts into a fresh temporary directory, and runs the frozen component replayers without editing their source. The quick command, omitting `--full`, checks the final integration and its mutation tests. Full mode additionally regenerates R0’s finite identities, R1’s exact witnesses, the local-tail symbolic certificate, and the global matching and root certificates. No network access is required. The exact command scope and environment are recorded in `REPRODUCIBILITY.md`.

The fresh assembly checks passed the final integration replay, including 12 distinct nested archive identities, 590 nested manifest entries, the typed identification chain, strict endpoint signs, and all 13 rational Newton intersections. The component replays for R0, R1, local matching, and global uniqueness also passed. R1 rejects 13 mathematical mutations. The integration mutation checker rejects missing interval premises, changed matching objects, branch substitution, false endpoint signs, invalid Newton centers and other specified corruptions. The scope is finite rejection testing, not a proof that every possible implementation defect is absent. Fresh logs and a receipt are included in the manuscript archive.

The arithmetic layers have different contracts. R0 and R1 use arbitrary precision integers and rational arithmetic. The local fixed-point and characteristic identities use exact symbolic operations, with frozen SymPy 1.14.0 and its bundled dependency. Global matching uses a small project-local real/complex interval and first-derivative implementation, with directed Decimal contexts at precision 90. A square root is widened from its correctly rounded nearest value and accepted only after the squared endpoints enclose the argument. Complex rectangles certify the chosen square-root branch and nonzero division denominators. Analytic remainder bounds are added before propagation; they are not inferred from rounding alone. The final enclosure arithmetic is independently checked using rational numbers.

The executable trust base includes the checker sources, Python and its integer, rational and Decimal implementations, parsing and filesystem behavior, the operating system and hardware. The stable-manifold, compactness, contraction and variational arguments are ordinary analytic proofs rather than executable proof-assistant terms. Hashes establish custody and exact version identity,

not mathematical correctness. Detailed identities and certificate paths are in Appendices B–D; full archive hashes are in Table 4.

The surviving global-uniqueness source has archive digest beginning `106adfdb56e`. The rejected alternative branch is not used. The distinction is enforced by the frozen integration’s closed archive allowlist and exact source pins. Replaying a branch with a numerically matching answer would not meet this contract.

10 External premises and source provenance

The remaining non-elementary external theorem premises are exactly Coladangelo v1 Theorem 7.1, the finite-projective tensor clause of Theorem 7.2, Proposition 10.9, and the standard finite-dimensional real-analytic stable/unstable manifold theorem. The first three are stated in External premise 3.2 and refer to one fixed v1 PDF, whose identity is recorded in Appendix E. They share proof ancestry: Theorem 7.2 uses the scalar-certificate theory, while Proposition 10.9 also uses the source’s finite-approximation and equality arguments. Multiple AI-generated branches that consume these premises are not independent confirmations of them.

The parameterization-method article [1] supplies a standard reference for analytic regularity, with the stable/unstable specialization described in its Theorem 1.1 discussion. The textbook [7] is a reference for the classical local theory. The frozen provenance audit inspected the analytic article; it verified the textbook’s publisher metadata but did not successfully obtain and inspect its full text. This is not represented as a new independent proof of the general theorem. Its concrete applicability here—analyticity, invertibility, hyperbolicity and direction counts—is established in Lemma 5.2.

Pauwels’s preprint and its associated formal development are comparison sources, not premises of the final theorem. In particular, their formal statements are not a Lean certification of the residual zero or of this high-precision enclosure. Pál–Vértési is historical context and construction ancestry. No reported floating-point optimum or conjectured optimality from that work is assumed.

The R1 candidates are the two specific rational objects released by Mghirbi in v2.0.0 [3]. Released-byte identity and exact validity have been independently certified. Complete numerical-generation ancestry before that release has not been reconstructed. This distinction matters: the proof of Proposition 3.1 accepts the explicit projectors and polynomial identity after exact checking, and does not need the original solver trajectory to be correct or recoverable. The lower decimal printed in the frozen companion preprint is inaccurate in its last reported digits; the rational witness and its recomputed interval in this paper are the authority. Later source versions do not replace the frozen upper certificate.

The source audit records other local textual errors, including a false sentence asserting that shifting an irreducible Jacobi matrix by a scalar identity makes all its entries positive. Such a shift cannot change the zero corner entries of a tridiagonal matrix. The actual weaker fact used here is positivity on a connected component, established by absolute-value maximization and zero propagation in Appendix A.2; no false source sentence is imported. Appendix E records this and the remaining citation caveats.

The zero-Douglas statement is precise: no Douglas/Apsiap theorem, data, code, root, witness, or PASS result is used as mathematical authority anywhere in the final chain. Historical mentions in provenance records are not theorem premises. The assembly used no project-specific material outside the supplied archive. Fresh web access was limited to bibliographic metadata checks; no proof step was imported from it.

11 Limitations, validation status and disclosure

The mathematical chain is closed relative to the explicit external premises and finite arithmetic contracts stated above. The frozen proof package survives internal clean reconstruction, independent adversarial checking of specified components, exact replay, and the project-level source-provenance audit. This describes the recorded scope of checking. It does not turn model agreement into independent mathematical evidence. External expert human validation of the new exact-value theorem has *not yet occurred*, and journal peer review of this new result has *not yet occurred*.

The final integrated adversarial-audit verdict is **OPEN-MINOR**. Its sole remaining mathematical-package issue is M001: the generic Decimal midpoint helper can return a point outside an arbitrarily narrow input interval. For example, at precision 90 its midpoint of the point interval $1 - 10^{-90}$ can be 1. The defect remains genuine and unpatched in the frozen source. The sealed arithmetic audit reports that all 22,074 actual midpoint returns in the submitted traces lay within their respective input intervals, and that no submitted load-bearing enclosure was invalidated. The assembly’s independent rational integration replay checked the 13 actual root-Newton centers. It did not repeat the historical 107,716,343-operation observer audit. Thus M001 is retained as an open, non-load-bearing implementation defect, not relabelled as nonexistent.

The project source-provenance verdict is **SOURCE-PROVENANCE-OPEN-MINOR**. It preserves the known source-text errors, M001, and the unresolved complete historical generation ancestry of the R1 witnesses. According to the frozen audits none of these remaining issues is load-bearing for the theorem. They nevertheless belong in the validation record and should be available to human referees.

The theorem concerns only the specified finite-dimensional projective tensor supremum. It makes no commuting=tensor assertion, uses no R2, and does not claim uniqueness of a physical optimizer or carrier, finite-dimensional attainment, or an elementary closed form for the value.

AI and computational assistance

The supplied record documents substantial AI and computational assistance in proof discovery, clean reconstruction, certificate generation, adversarial auditing and source-provenance checking. This manuscript’s exposition, notation normalization, supporting documentation and assembly were also prepared with AI assistance. Generated arguments and scouting numerics are distinguished from exact identity checking and rigorous interval certification. The fresh replays reported here run specified checkers against frozen inputs; their success is not a claim that every analytic argument or the software stack has been formally verified. Different AI runs may share ideas, data and external premises, so their agreement is not counted as independent mathematical evidence. External human expertise and peer review remain separate, pending forms of validation. No AI system is listed as an author. The responsible author must review the mathematics, citations, disclosure and final submission. Acknowledgements are deferred until external human review.

Funding

The author received no specific funding for this work.

Competing interests

The author declares no competing interests.

A The analytic outer-carrier construction

This appendix supplies the analytic details of Theorem 4.2. Throughout put $s = S_t = q^*$. Proposition 3.1 gives $1/4 < s < 13/50 < 1/3$. The only external I_{3322} premises are those in External premise 3.2.

A.1 Finite realization

For a path in (3.4), on the standard basis of $\mathbb{C}^m$ define diagonal operators by

$$((Z_A)_{ii}, (Z_B)_{ii}) = \begin{cases} (\gamma_{i-1}, \gamma_i), & i \text{ odd}, \\ (-\gamma_i, -\gamma_{i-1}), & i \text{ even}. \end{cases}$$

On even edges $(i, i+1)$, let X_A be the swap with weight $w(\gamma_i)$; on odd edges let X_B be the weighted swap. They vanish on unmatched endpoint basis vectors. Let W_B be the unweighted swap on even edges, and W_A the unweighted swap on odd edges, both with value $+1$ on unmatched vectors. The matched blocks satisfy $ZX + XZ = 0$ and $Z^2 + X^2 = I$. At an unmatched endpoint $X = 0$, $Z = \pm I$, because the endpoint label is signed. Consequently

$$A_1 = (I + Z_A - X_A)/2, \quad A_2 = (I + Z_A + X_A)/2, \quad A_3 = (I + W_A)/2$$

and

$$B_1 = (I + Z_B - X_B)/2, \quad B_2 = (I + Z_B + X_B)/2, \quad B_3 = (I + W_B)/2$$

are orthogonal projectors. Expansion gives

$$\mathcal{B} = Z_A Z_B + \frac{1}{2} Z_A - \frac{1}{2} Z_B + \frac{1}{2} X_A W_B + \frac{1}{2} W_A X_B - I.$$

For a real unit vector v , its expectation on $\sum_i v_i e_i \otimes e_i$ is $v^T J_m(\gamma) v$: the diagonal is $d(\gamma_{i-1}, \gamma_i)$ in either parity, and each edge contributes $w(\gamma_i) v_i v_{i+1}$. This proves the finite realization used in Proposition 3.3. The source's bilateral indexing is converted by $c_j = \gamma_{j-1}$, with the amplitude index unchanged. It gives exactly (3.6).

A.2 A nonzero square-summable limit

For any finite path and real f ,

$$\langle f, J_m(\gamma) f \rangle \leq \frac{1}{4} \|f\|_2^2 + \frac{1}{2} \text{TV}(\gamma) \|f\|_\infty^2. \quad (\text{A.1})$$

To see this, subtract $(x^2 - 1 + y^2 - 1)/2$ from $d(x, y)$: the result is $-(x - y)^2/2 + (x - y)/2 \leq |x - y|/2$. The remaining two-vertex form for an internal edge has matrix

$$\begin{pmatrix} -w^2/2 & w/2 \\ w/2 & -w^2/2 \end{pmatrix},$$

whose top eigenvalue is $-w^2/2 + w/2 \leq 1/8$. Summing counts each vertex at most twice; the signed endpoints contribute zero. This proves (A.1).

Take the monotone paths of External premise 3.2(3) and unit top eigenvectors $f^{(n)} \geq 0$. Absolute values cannot reduce the Rayleigh quotient when off-diagonals are nonnegative, so these nonnegative choices exist. Their values s_n tend to $s > 1/4$; since their variation is two, (A.1) implies $\|f^{(n)}\|_\infty^2 \geq s_n - 1/4$. Translate a maximum coordinate to zero. Extend the labels by $+1$ to the left and -1 to

the right, and amplitudes by zero. These are bilateral eigenvectors because the separating endpoint edges have zero off-diagonal entries.

A diagonal subsequence in the compact coordinate intervals gives nonincreasing labels $c_j \in [-1, 1]$ and amplitudes $f_j \geq 0$. Every finite sum of f_j^2 is at most one, so $f \in \ell^2(\mathbb{Z})$. The lower bound at zero makes $f \neq 0$. Continuity of b, d in each three-term equation gives $H(c)f = sf$. Set $\lambda = f/\|f\|_2$. At this point neither positivity at every site nor strict decrease has been assumed.

A.3 Boundary labels and support

At a zero amplitude, the nonnegative eigenvalue equation forces neighboring amplitudes across positive edges to vanish. Each connected component of positive edges is therefore either identically zero or strictly positive. Monotonicity of the labels leaves at most one component containing positive edges. Other components are isolated vertices with $d \leq 0 < s$, hence zero amplitudes. A nonzero component exists and contains an edge.

Suppose this component has a left endpoint j_0 . Then $c_{j_0-1} = c_{j_0} = 1$, $\lambda_{j_0-1} = 0$ and $\lambda_{j_0} > 0$. Choose

$$0 < \varepsilon < \min\{1, (1 - 3s)/2\}, \quad c'_{j_0} = 1 - \varepsilon, \quad f = \lambda + \alpha \lambda_{j_0} e_{j_0-1}, \quad \alpha = \frac{\sqrt{2\varepsilon - \varepsilon^2}}{2s + \varepsilon}.$$

A direct two-vertex computation gives

$$\frac{\langle f, H(c')f \rangle - s\|f\|_2^2}{\lambda_{j_0}^2} = \varepsilon \left[-(c_{j_0+1} + 1/2) + \frac{2 - \varepsilon}{4s + 2\varepsilon} \right] > 0.$$

The last fraction exceeds $3/2$, whereas $c_{j_0+1} + 1/2 \leq 3/2$. This contradicts the universal comparison in Proposition 3.3. At a right endpoint j_1 , change $c_{j_1+1} = -1$ to $-1 + \varepsilon$ and add $\alpha \lambda_{j_1} e_{j_1+1}$. The bracket becomes $-(1/2 - c_{j_1}) + (2 - \varepsilon)/(4s + 2\varepsilon) > 0$, the same contradiction. Thus the component is all of $\mathbb{Z}$:

$$-1 < c_j < 1, \quad \lambda_j > 0 \quad (j \in \mathbb{Z}). \quad (\text{A.2})$$

This argument establishes the weaker positivity fact actually needed; it does not use the false source sentence about all entries becoming positive after a scalar identity shift.

A.4 Finite label variations

For fixed λ , the Rayleigh form is

$$Q(c) = \sum_j d(c_j, c_{j+1}) \lambda_j^2 + \sum_j w(c_j) \lambda_{j-1} \lambda_j.$$

It is absolutely convergent by bounded coefficients and $\sum_j \lambda_{j-1} \lambda_j \leq 1$. A finite change of interior labels changes only finitely many smooth summands. The comparison $Q(c') \leq s$, with equality at c , makes the first derivative zero and the Hessian nonpositive for every finite variation. In particular

$$(c_{j-1} - 1/2) \lambda_{j-1}^2 + (c_{j+1} + 1/2) \lambda_j^2 - \frac{c_j}{w(c_j)} \lambda_{j-1} \lambda_j = 0 \quad (\text{A.3})$$

and

$$\frac{1}{2} D^2 Q[\delta, \delta] = -\frac{1}{2} \sum_j \frac{\lambda_{j-1} \lambda_j}{w(c_j)^3} \delta_j^2 + \sum_j \lambda_j^2 \delta_j \delta_{j+1} \leq 0. \quad (\text{A.4})$$

Together with the eigenvalue equation, (A.3) gives (4.5) and (4.3). All denominators are positive by (A.2).

A.5 Equality geometry and exclusion of plateaux

The function $\mathfrak{h}$ in (3.5) is bounded, since setting $x = 0$ gives $\mathfrak{h}(y) \leq s + 1 + y/2$; it is continuous in the interior by concavity. Absolute convergence allows the deficit identity

$$\begin{aligned} 0 &= s\|\lambda\|^2 - Q(c) \\ &= \sum_j [s - d(c_j, c_{j+1}) - \mathfrak{h}(c_{j+1}) - \mathfrak{h}(-c_j)]\lambda_j^2 \\ &\quad + \sum_j [\mathfrak{h}(c_j)\lambda_{j-1}^2 + \mathfrak{h}(-c_j)\lambda_j^2 - 2b(c_j)\lambda_{j-1}\lambda_j]. \end{aligned} \tag{A.5}$$

Every summand is nonnegative. For an edge, this follows by completing a square and using $\mathfrak{h}(c_j)\mathfrak{h}(-c_j) \geq b(c_j)^2$. Since all amplitudes are positive, equality forces

$$\begin{aligned} \mathfrak{h}(c_{j+1}) + d(c_j, c_{j+1}) + \mathfrak{h}(-c_j) &= s, \\ \mathfrak{h}(c_j)\mathfrak{h}(-c_j) &= b(c_j)^2, \quad \mathfrak{h}(c_j) = b(c_j)u_j. \end{aligned} \tag{A.6}$$

Here differentiability at saturated labels must be proved rather than assumed. If $t \in (-1, 1)$ and $\mathfrak{h}(t)\mathfrak{h}(-t) = b(t)^2$, both values of $\mathfrak{h}$ are positive. Concavity gives finite one-sided derivatives $\mathfrak{h}'_- \geq \mathfrak{h}'_+$ at both labels. For $p(z) = \mathfrak{h}(z)\mathfrak{h}(-z)$,

$$p'_-(t) = \mathfrak{h}'_-(t)\mathfrak{h}(-t) - \mathfrak{h}(t)\mathfrak{h}'_+(-t), \quad p'_+(t) = \mathfrak{h}'_+(t)\mathfrak{h}(-t) - \mathfrak{h}(t)\mathfrak{h}'_-(-t).$$

Thus $p'_- - p'_+$ is the sum of two nonnegative derivative jumps with positive weights. But $p - b^2 \geq 0$ has a zero at t , so $p'_-(t) \leq -t/2 \leq p'_+(t)$. All inequalities are equalities, and both jumps vanish. Consequently $\mathfrak{h}$ is differentiable at $t, -t$. Varying the two arguments at a saturated pair (x, y) gives

$$x = 1/2 - \mathfrak{h}'(y), \quad y = \mathfrak{h}'(-x) - 1/2. \tag{A.7}$$

These formulas determine the predecessor and successor uniquely among saturated equality pairs.

If $c_j = c_{j+1} = t$, (A.7) propagates this equality to each adjacent pair and hence to the entire sequence. For constant labels the quadratic form has upper bound

$$t^2 - 1 + w(t) = -w(t)^2 + w(t) \leq 1/4$$

on unit vectors, by Cauchy–Schwarz. This contradicts $s > 1/4$. Therefore $c_j > c_{j+1}$ everywhere.

A.6 Interior tail limits

Strict decrease gives distinct limits $t_-, t_+ \in [-1, 1]$. To exclude boundary limits before assuming ratio convergence, set $A_j = w(c_j)u_j$ and $B_j = w(c_j)/u_j$. Equation (A.3) and the Jacobi equation give

$$\begin{aligned} (c_{j+1} + 1/2)A_j + (c_{j-1} - 1/2)B_j &= c_j, & A_j B_j &= 1 - c_j^2, \\ B_j + A_{j+1} &= 2(s - d(c_j, c_{j+1})). \end{aligned} \tag{A.8}$$

If a tail tends to $+1$, the first equation bounds both positive quantities on that tail. Their possible subsequential limits are $(2/3, 0)$ and $(0, 2)$. If a tail tends to -1 , multiply the first equation by -1 ; the possible pairs are $(2, 0)$ and $(0, 2/3)$. Simultaneous convergent subsequences at neighboring indices, substituted in the second equation with $d \rightarrow 0$, would give $s \in \{0, 1/3, 1, 4/3\}$. This contradicts the initial interval. Hence both limits are interior, and the labels lie in a fixed compact subinterval of $(-1, 1)$.

Lemma A.1 (Positive Jacobi tail ratios). *Suppose a positive right-tail sequence satisfies*

$$a_n f_{n-1} + d_n f_n + a_{n+1} f_{n+1} = s f_n, \quad a_n \rightarrow a > 0, \quad d_n \rightarrow d, \quad K = (s - d)/a > 2.$$

If it is square summable, then

$$\frac{f_n}{f_{n-1}} \rightarrow \frac{K - \sqrt{K^2 - 4}}{2} < 1.$$

For a square-summable left tail the forward ratio tends to the reciprocal root, which exceeds one.

Proof. Positivity gives, for $v_n = f_n/f_{n-1}$,

$$v_n > \frac{a_n}{s - d_n}, \quad v_{n+1} < \frac{s - d_n}{a_{n+1}}.$$

Thus late ratios stay in a compact positive interval. The recurrence is $v_{n+1} = F(v_n) + o(1)$, uniformly there, with $F(v) = K - 1/v$. As F is increasing and continuous, the liminf and limsup are each fixed points of F . These are $\rho_- < 1 < \rho_+$. They cannot be different: for any $m \in (\rho_-, \rho_+)$, $F(m) > m$, and sufficiently small late errors make crossing above m irreversible. A limsup of ρ_+ would force such a crossing, contradicting liminf ρ_- . The ratio therefore converges. Square summability excludes ρ_+ , since ratios eventually bounded below by a constant greater than one force growth. Reverse the sequence for the left tail. $\square$

For a carrier tail with label limit t , the limiting coefficients are $a = w(t)/2$, $d = t^2 - 1$. The condition $K > 2$ follows from

$$s > 1/4 \geq t^2 - 1 + w(t).$$

Lemma A.1 gives full positive finite ratio limits $u_- > 1$ on the left and $u_+ < 1$ on the right.

A.7 Tail elimination and exclusion of the extraneous branch

Passing to the now justified limits in the two stationary equations, write the label and ratio as t, r , with $w = \sqrt{1 - t^2} > 0$:

$$\frac{w}{2}(r + r^{-1}) + t^2 - 1 = s, \quad (t + 1/2)r + (t - 1/2)r^{-1} = t/w. \quad (\text{A.9})$$

Putting $h_0 = s - t^2$ gives

$$r + r^{-1} = \frac{2(h_0 + 1)}{w}, \quad r - r^{-1} = -\frac{2t(2h_0 + 1)}{w}.$$

Their squared difference equals four, so

$$0 = (s - t^2)\{4t^4 - (4s + 5)t^2 + s + 2\}. \quad (\text{A.10})$$

The coefficient identity is checked exactly, without inserting a preferred root branch.

In (A.4) take a finite variation $\delta_j = z_j/\sqrt{\lambda_{j-1}\lambda_j}$. Its finitely many entries are finite even far out a small-amplitude tail, so a sufficiently small perturbation parameter keeps the labels interior. Twice the resulting form is

$$-\sum_j w(c_j)^{-3} z_j^2 + 2 \sum_j \sqrt{u_j/u_{j+1}} z_j z_{j+1} \leq 0.$$

Choose $z = 1$ on five consecutive sites and translate them down either tail. The form tends to $8 - 5/(1 - t^2)^{3/2}$. On $s = t^2$, this is strictly positive, since

$$(1 - s)^3 > (37/50)^3 > 25/64 = (5/8)^2.$$

The last comparison is exact rational arithmetic. Sufficiently far translated variations then contradict the nonpositive form. Hence the branch $s = t^2$ is excluded.

A.8 Outer rather than inner or mixed tails

The remaining polynomial $P_s(z) = 4z^2 - (4s + 5)z + s + 2$ satisfies

$$P_s(1/4) = 1, \quad P_s(3/4) = 1/2 - 2s < 0, \quad P_s(1) = 1 - 3s > 0.$$

It has two roots with

$$1/4 < \beta(s)^2 < 3/4 < a(s)^2 < 1, \quad \beta(s) = \sqrt{\frac{4s + 5 - \sqrt{16s^2 + 24s - 7}}{8}}.$$

Adding the sum and difference equations above and using

$$(2t + 1)[(s + 1 - t^2) - t(2s + 1 - 2t^2)] - (1 + t)(2t - 1) = P_s(t^2)$$

gives $r = r(t)$ as in (4.2). All denominators are nonzero because the root magnitudes exceed $1/2$. Moreover

$$r(t)^2 - 1 = \frac{2t(4t^2 - 3)}{(1 - t)(2t + 1)^2}, \quad r(-t) = r(t)^{-1}.$$

The left ratio > 1 permits labels $a, -\beta$, and the right ratio < 1 permits labels $-\alpha, \beta$.

Root ordering alone does not remove the same-sign pairs (a, β) and $(-\beta, -a)$. For this step use the scalar certificate. Taking limits in (A.6) gives

$$\mathfrak{h}(t) + \mathfrak{h}(-t) + t^2 - 1 = s, \quad \mathfrak{h}(t)\mathfrak{h}(-t) = b(t)^2, \quad \mathfrak{h}(t) = b(t)r(t).$$

The constant pair with both signs reversed also saturates (3.5). Differentiability and (A.7) then imply, at either positive magnitude $m = |t|$,

$$\mathfrak{h}'(m) = 1/2 - m, \quad \mathfrak{h}(m) = f_0(m) := \frac{(1 + m)(2m - 1)}{2(2m + 1)} = \frac{m}{2} - \frac{1}{2(2m + 1)}.$$

If both magnitudes $a > \beta > 1/2$ occurred, the supporting line at β would give

$$\mathfrak{h}(a) \leq \mathfrak{h}(\beta) + (1/2 - \beta)(a - \beta) < \mathfrak{h}(\beta).$$

But

$$f_0(a) - f_0(\beta) = (a - \beta) \left(\frac{1}{2} + \frac{1}{(2a + 1)(2\beta + 1)} \right) > 0,$$

a contradiction. Mixed magnitudes are impossible. Of the two remaining equal-magnitude pairs, $(-\beta, \beta)$ is increasing, so strict decrease leaves precisely $(a, -a)$. The already established ratio limits are consequently $r(a) > 1$ and $r(-a) < 1$. This proves all of (4.4) at $s = S_t$, completing Theorem 4.2.

B Finite rational bound certificates

The rational objects are `lower_d499.json` and `upper_level14.json` in the canonical R1 component. They originate in the immutable v2.0.0 release [3]; the original license and attribution travel inside the supplied input archive. The following specifies exactly what their independent checker proves.

B.1 Lower witness

The first object contains rational circle pairs (c_i, s_i) , $1 \leq i < 499$, and rational amplitudes λ_i , $1 \leq i \leq 499$. Set $c_0 = 1, c_{499} = -1$. The finite predicates are $c_i^2 + s_i^2 = 1$, $s_i > 0$, and $N = \sum_i \lambda_i^2 > 0$. Define

$$R_i^\pm = \frac{1}{2} \begin{pmatrix} 1 - c_i & \pm s_i \\ \pm s_i & 1 + c_i \end{pmatrix}, \quad T_i^\pm = \frac{1}{2} \begin{pmatrix} 1 + c_i & \pm s_i \\ \pm s_i & 1 - c_i \end{pmatrix}, \quad J = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

The projectors in the target labels are the following direct sums.

Operator	Two-dimensional blocks	Singleton
A_1, A_2	R_i^-, R_i^+ , even $i = 2, \dots, 498$	1 on e_1
A_3	J , odd $i = 1, \dots, 497$	1 on e_{499}
B_1, B_2	T_i^-, T_i^+ , odd $i = 1, \dots, 497$	0 on e_{499}
B_3	J , even $i = 2, \dots, 498$	1 on e_1

Every block is symmetric and idempotent by the circle relation. The checker additionally multiplies all six full sparse 499×499 matrices, checking symmetry and idempotency at every entry. The state is $N^{-1/2} \sum_i \lambda_i e_i \otimes e_i$. For real matrices its expectation satisfies

$$\langle P \otimes Q \rangle = N^{-1} \sum_{i,j} \lambda_i \lambda_j P_{ij} Q_{ij}, \quad \langle P \otimes I \rangle = N^{-1} \sum_i \lambda_i^2 P_{ii}.$$

Direct contraction of (2.1), including all zero coefficients, is compared with the tridiagonal formula. The result is

$$v = N^{-1} \left(\sum_i d(c_{i-1}, c_i) \lambda_i^2 + \sum_{i=1}^{498} s_i \lambda_i \lambda_{i+1} \right).$$

The reduced numerator and denominator have 5,851 and 5,852 decimal digits and are stored in `certificate/lower_receipt.json`. Exact integer comparison, without a square-root approximation to the state normalization, proves the interval for v in Proposition 3.1.

B.2 Upper polynomial identity

Generate all reduced single-party words in three involutions up to length four, pair Alice and Bob words with total length at most four, and order the 244 pairs by length and then lexicographically. Adjacent equal letters cancel, and cross-party letters commute. Adjoints reverse the order within each party. These reductions are identities in every allowed operator representation.

On the ordered generator tuple $(X_1, X_2, X_3, Y_1, Y_2, Y_3)$, use the signed substitutions

$$r = (-Y_1, -Y_2, Y_3, -X_2, -X_1, -X_3), \quad s_0 = (X_1, X_2, -X_3, Y_2, Y_1, Y_3).$$

For each word-basis column and for signs $(a, b) = (1, 1), (1, -1), (-1, 1), (-1, -1)$, form $\sum_{k=0}^3 a^k (r^k + b s_0 r^k)$ applied to that column. Select the first linearly independent columns using rational elimination, divide each by the positive gcd of its integer entries, and preserve its sign. This constructs $U_1, \dots, U_4$ with 31, 30, 26 and 35 columns. The same operation on $(1 + s_0)(1 - r^2)$ gives U_5 with 61 columns; set $U_6 = r U_5$. The construction only defines concrete integer matrices. No assertion that they exhaust a symmetry decomposition is used.

The candidate gives five symmetric integer matrices M_k of sizes 31, 30, 26, 35 and 61, with common denominator

$$D_0 = 18534051411918436794453781905408000000000 > 0.$$

Exact symmetric fraction-free elimination checks positive leading determinants Δ_j , with $\Delta_0 = 1$, using the update

$$b'_{ij} = \frac{b_{kk}b_{ij} - b_{ik}b_{kj}}{\Delta_{k-1}}.$$

Each division must have positive denominator and zero remainder. The Schur-complement determinant identity shows inductively that the pivots are leading principal determinants; the corresponding congruent diagonal entries are $\Delta_j/\Delta_{j-1} > 0$. Thus all five M_k are positive definite. The certificate stores all 183 determinants.

Construct

$$Y = D_0^{-1} \left(\sum_{k=1}^4 U_k M_k U_k^T + U_5 M_5 U_5^T + U_6 M_5 U_6^T \right) \succeq 0.$$

Real positivity follows term by term under congruence, and complex positivity by real and imaginary parts. The full Gram numerator is reconstructed in `certificate/reconstructed_gram.json`. Expansion of every ordered product $w_i^* w_j$, reduction as above, and comparison at all 8,452 normal-form positions prove (3.2). To pass from the finite identity to an operator inequality, evaluate the words on a strategy and set $\eta = (w_i \xi)_i$. Then $\langle \xi, \mathbf{w}^* Y \mathbf{w} \xi \rangle = \eta^* (Y \otimes I) \eta \geq 0$. This proves the uniform upper bound for every finite dimension.

B.3 Acceptance and mutation contract

Both candidates are parsed with duplicate-key and nonfinite-number rejection; rational fields reject floating values and invalid denominators. The normalization coefficient dictionaries must agree exactly, the deterministic local maximum must be zero, all lower projector and value checks must pass, all upper determinant and coefficient checks must pass, and the rational endpoint comparisons must hold. Each condition fails by an explicit exception. Fresh receipts are compared with the frozen ones only after recomputation. R1's 13 mutation probes test mathematical rejection paths; original candidate generation is not rerun and is not a proof premise.

C Symbolic identities and interval certificate specifications

C.1 Exact algebra

All rational-map identities in Lemma 6.1 are tested by clearing denominators on their stated nonzero domain and requiring the numerator to be the zero polynomial. Sparse coefficients are exact rationals. The checker also embeds the production propagation and Jacobian formulas in the independent polynomial arithmetic and compares each entry with direct differentiation. Pointwise numerical agreement is not the acceptance criterion.

In canonical coordinates the local symbolic certificate supplies the six fixed-point residuals, the two characteristic polynomials and Jacobian determinants, the reciprocal-spectrum identities, $r(t)r(-t) = 1$, the parameter inverse polynomial and σ' . The characteristic sign argument in Lemma 5.2 then establishes the relevant dimensions; it is not a floating-point eigenvalue count. The R0 finite checker separately regenerates 35 polynomial identities and 15 rational comparisons supporting Appendix A.

For the rational-coordinate proof there are 48 exact identities. Their roles are the implemented-map and Jacobian comparison, canonical conjugacy, both inverse identities, reverser involution and reversibility, contact covariance and determinant, fixed-point and spectral formulas, the tangent

eigenvector, projected contact differentiation and the conditional scalar derivative cancellation. One useful propagated phase identity, obtained by differentiating (6.2) and substituting (6.8), is

$$z_\zeta = -\frac{B_0(y)}{q^2}x_\zeta + \left(\frac{1}{2q} + \frac{(x-1/2)y}{q^2} - \frac{2(x-1/2)^2B_0(y)}{q^3}\right)y_\zeta,$$

$$Q_\zeta = (1/2 - y)z_\zeta.$$

The independently differentiated formula must agree exactly before its interval version is used to tighten a propagated derivative enclosure. The identities alone do not assert completeness of a matching chart, symmetry of a full match, or a derivative sign.

C.2 Exact interval domain and analytic majorants

The outer interval $T^\square = [t_-^\square, t_+^\square]$ is the outward evaluation of the two radicals $a(L), a(U)$ by the frozen interval code. Its exact rational decimal endpoints are

$$t_-^\square = 0.878272945180812199461677854479963547717241304597573912661746000456489010475417113206932884$$

$$t_+^\square = 0.878272952173734583819480075317750711672962664979312588545615521238490507114264476586355651$$

The local complex rectangle is $\{t : |\Re t - t_c| \leq 10^{-5}, |\Im t| \leq 10^{-5}\}$, where t_c is the recorded real center. Its positive distance from $T^\square$ is Δ_t , used in the Cauchy derivative error. All coefficients are enclosed on this complex rectangle, rather than only at real sample points.

For completeness the majorants used in Certificate 6.2 can be checked from the coefficient boxes as follows. For a phase radius h_0 and correction allowance e_0 , set

$$d_i(h_0) = \sum_{n=1}^N |c_{n,i}| h_0^n, \quad X = |t| + d_x(h_0) + e_0, \quad Y = |t| + d_y(h_0) + e_0,$$

and let $q_0 = \inf |G(t)| - d_q(h_0) - e_0 > 0$. Put

$$B_M = (1 + Y^2)/4, \quad Z = \frac{Y}{2q_0} + \frac{(X + 1/2)B_M}{q_0^2} + \frac{1}{2}.$$

The coordinate bounds for $\hat{F}_{\sigma(t)}$ are

$$Y, \quad Z, \quad |\sigma(t)| + YZ + (Y + Z)/2 + 1 + B_M/q_0.$$

For its Jacobian, define

$$a_1 = \frac{B_M}{q_0^2}, \quad a_2 = \frac{1}{2q_0} + \frac{(X + 1/2)Y}{2q_0^2}, \quad a_3 = \frac{Y}{2q_0^2} + \frac{2(X + 1/2)B_M}{q_0^3}.$$

An upper row-sum bound is

$$L_F = \max \left\{ 1, a_1 + a_2 + a_3, (Y + 1/2)(a_1 + a_2 + a_3) + Z + 1/2 + \frac{Y}{2q_0} + \frac{B_M}{q_0^2} \right\}.$$

Use $h_0 = R_0/m, e_0 = 0$ to bound the defect on $R_0 = 0.2$ by the sum of the propagation and polynomial bounds. Use $h_0 = r_0/m, e_0 = \epsilon_N/m^{N+1}$ for the contraction bound on $r_0 = 0.01$. Complex divisions and square roots reject a rectangle meeting a forbidden denominator or branch. The finite gates are $m > 1$, $q_0 > 0$, $\Delta_t > 0$, $\kappa < 1/2$, and $\delta_N + \kappa\epsilon_N \leq \epsilon_N$, with the quantities defined in the proof of Certificate 6.2. These finite inequalities are what turn the polynomial into an analytic enclosure.

The recorded corrections and contractions admit the following outward rational summaries, independently compared with the exact receipts:

Quantity	Degree 36	Degree 48
Correction bound	$< 3.414 \cdot 10^{-47}$	$< 8.334 \cdot 10^{-63}$
Contraction factor	$< 6.891 \cdot 10^{-27}$	$< 4.409 \cdot 10^{-36}$
Complex $ q $ lower bound	> 0.2444	> 0.2444
$ \mu $ lower bound	> 5.8365	> 5.8365
Parameter margin	$> 9.996 \cdot 10^{-6}$	$> 9.996 \cdot 10^{-6}$

C.3 Cover and scalar receipts

Each geometry-tree node records a closed phase interval. A nonterminal node must have two children whose outer endpoints equal the parent's and whose inner endpoints agree exactly. Both children must be nondegenerate. Every node must be reachable once from the root; cycles, missing children and unused nodes are rejected. A geometry leaf records, for every iterate, the three state intervals, three phase-derivative intervals and the complementary- q interval. Strict endpoint signs certify the predicates of Certificate 6.3.

Every matching-cover leaf covers $T^\square$, its source-phase cell, and *all* possible target phases in the original envelope. The projected target interval contains every solution of the monotone inverse equation. A leaf records either the outside-section label intervals or the projected $\mathcal{H}, \mathcal{K}, \mathcal{H}_\zeta$ intervals and associated derivatives. The accepted terminal predicates are precisely those in Certificate 6.5. The separate central tree covers all of C_0 , including cells where the main tree used another predicate. This is why global central monotonicity does not follow merely from the two retained central leaves.

The scalar receipt encloses the whole implicit phase family inside

$$[0.0005824795297982988, 0.0005824795369892252].$$

The exact derivative interval is

lower endpoint:

3.48002909287387920997544898183182987998806066768305662514348825306622243384968318433976825

upper endpoint:

3.57068402546763126912045926632115988647904165952465340467067063466386183321219918274406572

It lies strictly in $(3.4800, 3.5707)$. The receipt also encloses σ' by positive endpoints, computes the quotient derivative in s , and verifies the two global endpoint residual signs. The root receipt records each input interval, exact center, residual interval and output interval in (8.1). Independent rational arithmetic checks the quotient enclosure, all 13 intersections, their nesting and strict shrinkage, the final 10^{-52} inflation, the strict signs and exact width.

C.4 Source-level certificate locator

Paths in this table are relative to the component roots identified in Appendix D; they are available offline in the bundled input archive.

Component	Certificate or checker	Mathematical output
R0	<code>certificates/finite_exact.json</code> ; <code>src/exact_checks.py</code>	Tail polynomial identities, finite comparisons, root signs; analytic construction remains Appendix A.
R1	<code>certificate/candidate/</code> ; <code>src/lower.py</code> , <code>src/upper.py</code>	Valid finite strategy and positive exact word identity.

Local matching	<code>certificates/symbolic_certificate.json;</code> <code>src/symbolic_cert.py</code>	Fixed points, characteristic factors, inverse parameter formula.
Global matching	<code>certificates/symbolic.json;</code> <code>src/exact.py</code>	48 rational identities, including production-formula agreement.
Global matching	<code>certificates/local_manifold.json;</code> <code>certificates/local_manifold_refined.json;</code> <code>src/manifold.py</code>	Complex analytic-germ and derivative error bounds, degrees 36/48.
Global matching	<code>certificates/phase_geometry.json;</code> <code>certificates/full_cover.json;</code> <code>certificates/central_monotonicity.json;</code> <code>src/global_cover.py</code>	Complete valid chart; projected inverse; all off-central exclusions and full central monotonicity.
Global matching	<code>certificates/scalar.json;</code> <code>certificates/root.json;</code> <code>src/scalar.py</code>	Full-domain derivative, existence signs, 13-step strict bracket.
Final closure	<code>replay_all.py;</code> <code>logs/08_EXACT_CHECKS.json</code>	Frozen-interface composition and rational integration arithmetic.

D Conventions, dependencies and frozen-source crosswalk

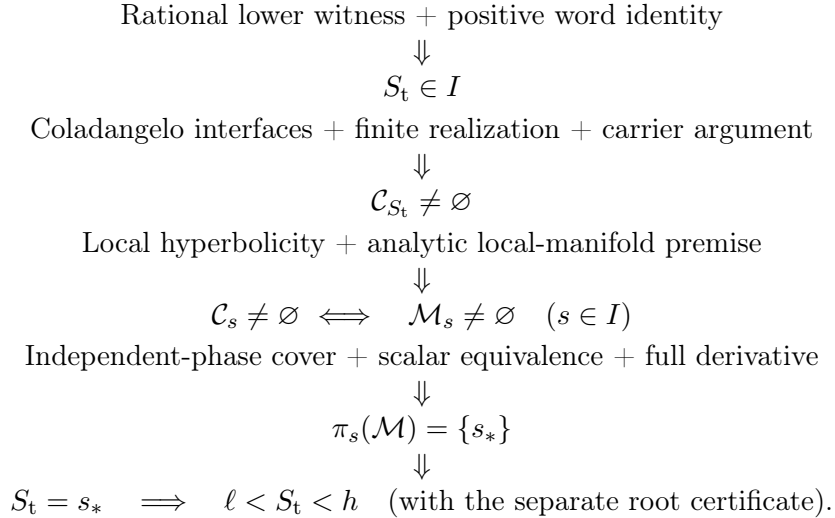
D.1 Notation crosswalk

This manuscript	Frozen convention	Meaning
S_t	S_t	Finite-dimensional projective tensor supremum; the subscript is not a variable.
$s, t = a(s)$	Same	Eigenvalue/Bell parameter and positive outer label, with $s = \sigma(t)$.
$b(c)$	Coladangelo's $a(c)$	Jacobi off-diagonal $\sqrt{1 - c^2}/2$, distinct from the outer magnitude $a(s)$.
c_j	Source γ_{j-1}	Label index shifted by one; amplitude index unchanged.
u_j	Canonical u_j	λ_j/λ_{j-1} .
ζ, η	Global component u, v	Independent tail phases, not canonical amplitude ratios.
$q, \bar{q}$	Global component q, h	$q = b(c_{j+1})\lambda_{j+1}/\lambda_j$ and $s - d - q$.
$\mathfrak{h}$	Upstream h	External concave scalar certificate, distinct from the upper rational endpoint h .
$\mathcal{H}, \mathcal{K}$	Global component H, K	Projected matching residuals; $H(c)$ remains the Jacobi operator.
$B_0(c)$	Global component $B(c)$	$b(c)^2 = (1 - c^2)/4$; $\mathcal{B}$ remains the Bell operator.

C_0	Global component C	Central phase interval; $C(t)$ in (6.6) is a spectral polynomial.
$\mathcal{R}$	Rcal	Exact analytic scalar residual after global asymmetric exclusion.

D.2 Dependency graph

The load-bearing logical graph is



The analytic local-manifold premise enters both the matching bridge and the identification of the certified analytic germ with the intrinsic tail. The global uniqueness theorem consumes the entire local-matching interface: full-state coordinates, valid chains, neighborhood independence and exact-zero ownership. Audit and custody records are validation nodes, not replacement theorem premises.

D.3 Obligation-to-manuscript crosswalk

Frozen label	Manuscript statement	Canonical source
R1	Proposition 3.1, Appendix B	Upstream R1 PROOF.md, normalization and exact candidate checkers.
R0	Theorem 4.2, Appendix A	Upstream R0 PROOF.md, all fourteen analytic steps.
O1	(3.1), (5.1)	Local component 01_PARAMETER_DOMAIN.md, DEFINITIONS.md D1.
O2	Lemma 5.2 and neighborhood independence	02_LOCAL_TAILS.md, 01_05_THEOREM.md items 1–2.
O3	Proposition 5.4, soundness direction	03_MATCHING_SOUNDNESS.md; three-coordinate concatenation, positive normalized square-summable amplitudes.
O4	Proposition 5.4, completeness direction	04_CARRIER_COMPLETENESS.md; arbitrary carrier, finite entry, crossing beginning at an exact zero.

O5	Proposition 5.5	O5_MATCH_EXISTENCE.md; R1 domain and existential R0.
O6	Lemma 6.1 through Theorem 6.9	Surviving PROOF.md, symbolic, local-manifold, global-phase, asymmetric, scalar and uniqueness components.
O7	Theorem 7.1	Final closure O7_TRUE_VALUE_IDENTIFICATION.md.
O8	Theorem 8.1	Final closure O8_ROOT_CERTIFICATION.md and surviving ROOT_ISOLATION.md, certificates/root.json.

D.4 Custody and offline locations

The original assembly archive has a single root I3322_PAPER_ASSEMBLY_INPUTS/. Its three numbered archives are the final clean closure, final integrated audit and source-provenance audit. Inside the closure’s inputs/ directory, component A is 01_CANONICAL_CLEAN_UPSTREAM_76d7.zip; B is 02_01_05_CANONICAL_PROOF_b7fc.zip; C and D are the local-component audit and provenance supplement; E is 05_06_SURVIVOR_PROOF_106ad.zip; F is 06_06_FINAL_AUDIT_CONSOLIDATION_18a9.zip. Inside A, the directories R0_CANONICAL/ and R1_CANONICAL/ contain the two upstream proof archives.

The upstream interface in A and its copies in B and E’s embedded conditional handoff are byte-identical. The local-matching definitions in B and E’s handoff differ in prose; their correspondence is semantic: the same $I, F_s, H, a, r, \mathcal{D}, \Sigma$, full-state chains, independent tail points/counts, carrier class and local-capture statements are used. The closure’s COMPONENT_INTERFACES.md records each correspondence. No generic text parser is claimed to prove this semantic identity.

Table 4: Frozen identities. Concatenate the two digest lines in each row.

Object	SHA-256
Original assembly input	73991d0bf7417de9586fbaf4f8649005 62f67f535732caa10299b374c7b3aa10
Final clean closure	e592932a0e12ee7fa97d9a1a30b84abd 17a3293f1546fda9fd9818415c34edea
Final integrated audit	7b576bff6886029e1e00cbf9f70c51be 42f50c2076c2af65cf6fab6843db24ef
Source-provenance audit	b517f69ecfd22084ffcc51251be928a1 340432157bc4359e85f1fa9156ff063c
A: canonical upstream	76d7f205d494d09c791d4a862794b6ef 5603cc02c494bddf0efb4368ad7ec052
B: canonical local matching	b7fc43c9f2ad219235e6dfe6d63fe803 0f308cb1b6ad56263d2f91283d86989f
E: surviving global matching	106adfddeb56eb792579a7dd0ab9de83d fd06dfe3f4abfcf7b1997abe5ea113ae
F: global audit consolidation	18a92a9ddaaa326eff3ce71588e41256 18108c9ec4d0bed4430d94104b6aad12
E: exact root certificate	6a0154a67ec857c2565dd5036745cd59 1658f6dea668b6cb3d03f84d8b062117

R1: lower rational candidate	67a06b7da3f40085603a4e52aa8c0e3e c3587f1665d2f884276e22631455d802
R1: upper rational candidate	a19b1d27b157d7acd76f39f5cc52f958 2242646e6386bc8dd8849e465be6119e

E Audit status and source caveats

Item	Frozen disposition and interpretation
R0, R1	CLOSED-PASS ; scoped analytic and finite components survive their checks.
Local matching (O1–O5)	CLOSED-PASS after the supplied provenance supplement; analytic local-manifold premise remains explicit.
Global uniqueness (O6)	Theorem branch survives; OPEN-MINOR because M001 remains genuine and non-load-bearing.
Final integrated audit	OPEN-MINOR ; final exact-value theorem survives. M001 is the sole remaining mathematical-package issue.
Source-provenance audit	SOURCE-PROVENANCE-OPEN-MINOR ; no load-bearing source error reported.
R1 witness provenance	Released identity and independent exact validity certified; complete earlier numerical-generation ancestry unresolved.
External expert human validation	Not yet completed for the new exact-value theorem.
Journal peer review	Not yet completed for the new exact-value theorem.

The final audit’s defect M001 is described in Section 11; its classification is retained without repair. A successful replay of the submitted path is compatible with a defective generic midpoint helper. It does not justify upgrading the overall verdict to a zero-defect status.

The source-provenance record lists six source-text errors and two additional caveats. Their relevant dispositions are:

1. Coladangelo v1 p. 22, Lemma 6.2: shifting by a scalar identity does not make every entry of a tridiagonal matrix positive. Use the actual positive-eigenvector and zero-propagation arguments; Appendix A.2 supplies the required weaker result.
2. Coladangelo v1 p. 19: a unit-circle sentence repeats c_i^2 . The correct relation is $c_i^2 + s_i^2 = 1$, used in the displayed blocks and the internal finite realization.
3. Coladangelo v1 p. 27, (43): the left-hand trace-norm bars are missing. The scalar trace-norm estimate actually established in (42) is the consumed statement; no stronger operator inequality is imported.
4. Coladangelo v1 p. 23: the local stationarity reference is (29)–(30), not (41).
5. Coladangelo v1 p. 35, Lemma 7.7: the pairwise condition used in the concave-envelope summation is (51), not (50).
6. The frozen Mghirbi preprint’s lower decimal 0.2508753845139766... disagrees with the exact released rational witness. Proposition 3.1 uses the independently recomputed interval, strictly above L .

7. Complete pre-release R1 generation ancestry remains unresolved. It is discovery history, not an unverified polynomial-identity premise.
8. M001 persists in the frozen O6 implementation and is not silently repaired by manuscript assembly.

The external Coladangelo premises refer to arXiv:2609.06038v1, whose frozen PDF SHA-256 is

596037dfb0c42ea5cda606aeb9764fd4aae
01dff7dbc24af0c43562cc1fa09dd

where the two lines are concatenated. Bibliographic metadata identifies Coladangelo and Pauwels as preprints in the checked records, not journal-accepted exact-value results. The Pál–Vértesi entry cites the APS version of record and its published title, rather than substituting the differently titled arXiv version. The Mghirbi manuscript and software release are distinct Zenodo records. The source-provenance audit’s publication check is dated 19 September 2026; the assembly’s limited fresh bibliographic checks are recorded in `SOURCE_PROVENANCE_NOTES.md`.

The historical arithmetic observer record totals 107,716,343 exact comparisons, 22,074 midpoint returns and 877 square roots, with zero failures on the submitted arithmetic paths. Its older raw observer archives are not supplied separately. This paper reports those figures as authenticated prior audit evidence, not as operations rerun during manuscript assembly. The fresh assembly record instead identifies the replays actually performed.

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